N18 Hamilton-Jacobi equations

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Consider a physical trajectory γ in extended phase space, inhabited by $\mathbf{r} = (q, p, t), \mathbf{u} = \dot{\mathbf{r}}$ and another trajectory $\tilde{\gamma}$ that differs to γ in initial momentum and final position, momentum and time. Connect the two trajectories by by two small path α and α' at the start and end respectively, the direction is for both α and α' is $\gamma \to \tilde{\gamma}$. The small ribbon is formed with surface Γ with boundary $\partial\Gamma = \gamma + \alpha' - \tilde{\gamma} - \alpha$. Classical dynamics satisfy

$$\nabla \cdot \mathbf{u} = 0 \tag{1}$$

for which we associate a vector potential $\boldsymbol{\theta} = (p, 0, -H)$, so that

$$\mathbf{u} = -\nabla \times \boldsymbol{\theta} \tag{2}$$

Then by Stokes' theorem

$$\oint_{\partial\Gamma} \boldsymbol{\theta} \cdot d\mathbf{r} = \iint_{\Gamma} \nabla \times \boldsymbol{\theta} \cdot d\boldsymbol{\sigma} \tag{3}$$

since the trajectory is along $\mathbf{u}$, we have $\mathbf{u} \cdot d\boldsymbol{\sigma} = 0$, thus

$$0 = \oint_{\partial\Gamma} \boldsymbol{\theta} \cdot d\mathbf{r} \tag{4}$$

then define action of a path

$$S = \int_{\gamma} \boldsymbol{\theta} \cdot d\mathbf{r} \tag{5}$$

Eq.(4) becomes

$$0 = S_{\gamma} + \int_{\alpha'} (p\, dq + H\, dt) - S_{\tilde{\gamma}} - \int_{\alpha} (p\, dq + H\, dt) \tag{6}$$

but $\tilde{\gamma}$ is only slightly different from γ, so we expect $S_{\tilde{\gamma}} = S_{\gamma} + dS$, and since the two trajectory start the same position and time, the integral over α vanishes, the integral over α' just becomes the small difference at the end (q', p', t), so

$$dS = p'\, dq' - H(q', p', t)\, dt \tag{7}$$

but we can also regard S as a function of q' and t, so we have

$$dS = \frac{\partial S}{\partial p'} dp' + \frac{\partial S}{\partial t} dt \tag{8}$$

comparing coefficients yields

$$\frac{\partial S}{\partial q'} = p', \qquad \frac{\partial S}{\partial t} = -H(q', p', t) \tag{9}$$

so momentum measures how action changes as we vary position, and energy measures how action changes as we delay the arrival. Eliminating p' gives the Hamilton-Jacobi equation

$$\boxed{\frac{\partial S}{\partial t} + H\left(q', \frac{\partial S}{\partial q'}, t\right) = 0} \tag{10}$$

for a typical $H = p^2/2m + V(q)$, it becomes

$$\frac{\partial S}{\partial t} + \frac{1}{2m}\left(\frac{\partial S}{\partial q'}\right)^2 + V(q') = 0. \tag{11}$$

We now make a connection to quantum mechanics, the time-dependent Schrödinger equation (TDSE) in position representation is

$$i\hbar\frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2 \psi}{\partial q^2} + V(q)\psi \tag{12}$$

the wavefunction have real amplitude $A(q,t)$ and real $S(q,t)/\hbar$

$$\psi(q,t) = A(q,t)e^{iS(q,t)/\hbar} \tag{13}$$

then substitute back into TDSE and doing the differentiation, then cancel the common phase factor of $e^{iS/\hbar}$ on both sides give

$$i\hbar\frac{\partial A}{\partial t} - A\frac{\partial S}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2 A}{\partial q^2} - \frac{i\hbar}{m}\frac{\partial A}{\partial q}\frac{\partial S}{\partial q} - \frac{i\hbar}{2m}A\frac{\partial^2 S}{\partial q^2} + \frac{A}{2m}\left(\frac{\partial S}{\partial q}\right)^2 + VA \tag{14}$$

comparing the imaginary part gives the time evolution of amplitude

$$\frac{\partial A}{\partial t} + \frac{1}{m}\frac{\partial A}{\partial q}\frac{\partial S}{\partial q} + \frac{A}{2m}\frac{\partial^2 S}{\partial q^2} = 0 \tag{15}$$

comparing the real part gives

$$\boxed{\frac{\partial S}{\partial t} + \frac{1}{2m}\left(\frac{\partial S}{\partial q}\right)^2 + V - \frac{\hbar^2}{2mA}\frac{\partial^2 A}{\partial q^2} = 0} \tag{16}$$

so the phase of a wavefunction has intimate connection with action, and Eq.(16) is the Hamilton Jacobi equation Eq.(10) with an addition quantum potential

$$Q = -\frac{\hbar^2}{2mA}\frac{\partial^2 A}{\partial q^2} \tag{17}$$

and the phase gradient is the classical momentum. Multiply Eq.(15) by $2A$, define probability density $\rho = A^2$ and rewrite as

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial q}\left(\frac{\rho}{m}\frac{\partial S}{\partial q}\right) = 0 \tag{18}$$

this is a continuity equation for probability conservation, with probability current

$$j = \frac{\rho}{m}\frac{\partial S}{\partial q} \tag{19}$$

so the Schrödinger equation is both an equation of amplitude, which is a continuity equation, an equation for phase, which is Hamilton-Jacobi equation, with a quantum correction potential Q.